

Central and Peripheral Fatigue: Interaction during Cycling Exercise in Humans

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ABSTRACT

AMANN, M. Central and Peripheral Fatigue: Interaction during Cycling Exercise in Humans. *Med. Sci. Sports Exerc.*, Vol. 43, No. 11, pp. 2039–2045, 2011. Existing evidence suggests that exercise-induced alterations of the metabolic milieu of locomotor muscle and associated peripheral muscle fatigue affect the central projection of thin-fiber muscle afferents. These neurons provide inhibitory feedback to the CNS and thereby influence the magnitude of central motor drive during high-intensity whole-body endurance exercise. The purpose of this proposed feedback loop would be to regulate and restrict the development of exercise-induced peripheral muscle fatigue and/or associated sensory feedback to an “individual critical threshold.” This centrally mediated restriction in the development of peripheral locomotor muscle fatigue might thereby help to prevent excessive disturbance of muscle homeostasis and potential harm to the organism. It seems that the regulatory mechanism is dominant during exercise under “normal” conditions but might become secondary in the face of extreme environmental influences such as severe hypoxia or heat. Most recent data are used to emphasize how the proposed feedback loop might be a key factor limiting performance during high-intensity whole-body endurance exercise. **Key Words:** EXERCISE LIMITATION, PERFORMANCE, MUSCLE AFFERENTS, INHIBITORY FEEDBACK

During strenuous exercise, the force/power-generating capacity of working skeletal muscle progressively declines; that is, fatigue develops until the task is terminated. This exercise-induced reduction of a muscle's ability to generate force/power is determined by a “peripheral” and/or a “central” component (1,19). The first comprises biochemical changes within the metabolic milieu of the working muscle leading to an attenuated response to neural excitation; the later comprises a failure of the CNS to “drive” the motor neurons, i.e., a reduction in central motor drive (CMD). The development of central fatigue during maximal isometric contractions of a single muscle has been linked with the central projection of group III and IV muscle afferents (19). The central projection of

these thin-fiber muscle afferents (i.e., relating “news” to the CNS regarding the status of the muscle) increases at the onset of exercise, at which contraction-induced mechanical and chemical stimuli begin to activate intramuscular receptors located at the terminal end of these sensory neurons (24,26).

There are numerous methods to assess exercise-induced peripheral locomotor muscle fatigue (13). In most of our investigations, we use supramaximal magnetic femoral nerve stimulation to evoke quadriceps twitch forces before and again immediately after exercise. Single¹ and/or paired stimuli are applied, and the decrease in evoked twitch forces from before to after exercise is used to quantify exercise-induced peripheral muscle fatigue.

DEVELOPMENT OF A HYPOTHESIS

The results from several studies during the past years reveal that the voluntary termination of exercise (i.e., exhaustion) or the end of a time trial task after high-intensity whole-body endurance exercise often coincides with a very

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¹Exercise-induced decreases in Ca²⁺ sensitivity and/or decreases in the amount of Ca²⁺ released from the sarcoplasmic reticulum during exercise may result in an overestimation of peripheral muscle fatigue when relying on the exercise-induced reductions in single-twitch forces versus tetanic forces (1).

specific and severe degree of peripheral locomotor muscle fatigue, a level that seems to be never exceeded voluntarily (3,5,7,10,15,17,33,34). On the basis of these findings, we proposed the existence of an “individual critical threshold” of peripheral locomotor muscle fatigue, which is associated with a certain sensory perception/degree of afferent feedback (5). The extent of end-exercise peripheral fatigue, i.e., the critical threshold, is presumably task specific (28) and varies across humans (5). The existence of a critical threshold of peripheral fatigue is supported by studies quantifying the biochemical status (which determines peripheral fatigue and the magnitude of group III/IV-mediated afferent feedback) of the working muscle at exhaustion after intense exercise. Their findings reveal that the exercise-induced intramuscular level of certain metabolites known to cause peripheral fatigue (e.g., hydrogen ions, inorganic phosphates; [1]) is usually very similar at exhaustion independent of the exercise regimen and the rate of change of intramuscular metabolic perturbation (12,16,21,22,35,39).

So, is it simply just coincidence that peripheral locomotor muscle fatigue after exhaustive high-intensity endurance exercise never exceeds a certain degree, i.e., the individual critical threshold? Or does the CNS deliberately regulate and limit the development of peripheral locomotor muscle fatigue—maybe to avoid overstraining/overexertion and potentially long-lasting harmful consequences for the muscle? Interestingly, the level of peripheral fatigue incurred at exhaustion, i.e., the critical threshold, does not depict the muscles’ ultimate limit (32) suggesting that exercise is regulated to retain a muscular “reserve capacity”—even at exhaustion/the voluntary termination of exercise (29,30). But how does the CNS monitor or sense peripheral muscle fatigue and/or the rate of development? It is likely that it is not peripheral locomotor muscle fatigue *per se* that is monitored but presumably the associated (and likely preceding) biochemical changes within the working muscle and the affiliated sensory perception/afferent feedback (i.e., the ensemble input of thin-fiber muscle afferents to the CNS [2]). Metabosensitive group III/IV muscle afferents relate exercise-induced metabolic perturbations within the working and fatiguing muscle to the CNS (24,26), and this (inhibitory) neural feedback may cause reductions in CMD; in other words, it may contribute to the development of central fatigue during exercise (18,19,36).

We interpreted existing correlative evidence (3,5,7,10,15,17,33,34) to mean that humans never voluntarily perform high-intensity endurance exercise to a degree that would incur peripheral locomotor muscle fatigue and associated sensation/perception beyond their individual critical threshold (or sensory tolerance limit). In other words, peripheral fatigue and associated sensory feedback during exercise under “normal” conditions (i.e., other than a life-or-death situation) only develops up to a threshold unique for each individual. Accordingly, either endurance exercise is voluntarily terminated once this critical threshold has been reached, in case of constant-load trials, or the exercise intensity is re-

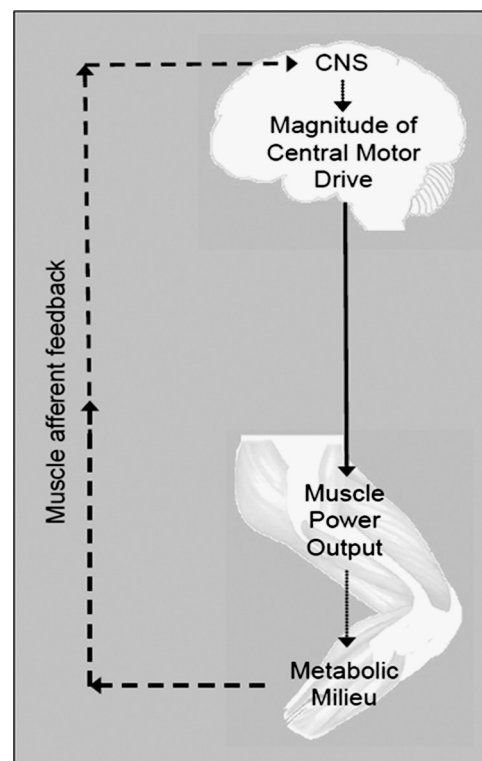


FIGURE 1—Schematic illustration of our working hypothesis. The solid line indicates CMD to the locomotor muscle; the dashed line indicates neural feedback mediated by thin-fiber muscle afferents. This regulatory mechanism suggests that muscle afferents exert inhibitory feedback effects on the determination of the magnitude of CMD during high-intensity whole-body endurance exercise. The magnitude of CMD determines power output of the locomotor muscles, which determines the metabolic milieu within the working muscles. The metabolic milieu determines the magnitude of the inhibitory afferent feedback. On the basis of existing data, this feedback loop restricts peripheral locomotor muscle fatigue and associated sensory feedback to an individual threshold and/or sensory tolerance limit that is never exceeded during whole-body endurance exercise. From Amann and Dempsey (4), used with permission.

duced (via reducing CMD) once a critical rate of fatigue development (or a critical rate of change in intramuscular metabolic milieu) is reached, in case of a time trial exercise.

We hypothesized that the CNS processes neural feedback from locomotor muscle afferents and regulates exercise by adjusting CMD to the locomotor muscle to confine/limit the development of peripheral fatigue to a critical threshold, beyond which the level of associated sensory input would not be tolerable (3–5,7–10) (Fig. 1). Stated differently, peripheral locomotor muscle fatigue and associated intramuscular metabolic changes exert, via the effects on lower limb muscle afferent feedback, an inhibitory influence on CMD and thus influence the development of central fatigue during high-intensity whole-body endurance exercise.

EXPERIMENTAL CHALLENGE OF HYPOTHESIS

We then used an interventional approach to directly test our hypothesis. Specifically, we asked if CMD—and

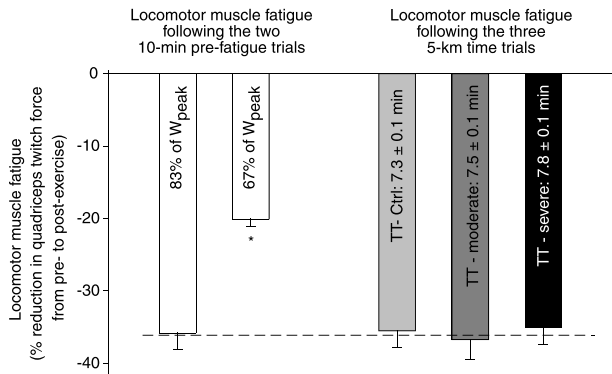


FIGURE 2—Locomotor muscle fatigue expressed as a percent change in quadriceps twitch force (magnetic femoral nerve stimulation) from before to 4 min after exercise. The two constant-workload trials (pre-fatigue trials: 83% of W_{peak} for 10 ± 1 min = 347 ± 14 W and 67% of W_{peak} for 10 ± 1 min = 276 ± 10 W) induced a severe and a moderate level of peripheral fatigue, respectively. The control time trial (TT-Ctrl) was conducted without preexisting locomotor muscle fatigue. The TT-moderate time trial was started 4 min after the 67% of W_{peak} pre-fatigue trial; the TT-severe time trial was started 4 min after the 83% of W_{peak} pre-fatigue trial. Note that despite significantly different levels of preexisting locomotor muscle fatigue, resulting in substantially different exercise performances, end-exercise locomotor muscle fatigue was almost identical between the three time trials (dashed line) supporting the hypothesis of an existing critical threshold of fatigue. $N = 8$. $*P < 0.01$. From Amann and Dempsey (3), used with permission.

therefore exercise performance—is regulated to avoid the development of peripheral locomotor muscle fatigue beyond an individual critical threshold. On two separate days, subjects performed constant-load cycling exercise; the first trial was performed to voluntary exhaustion at 83% of the subjects' peak power output (83% W_{peak} , ~347 W for ~10 min); the second trial was performed for the identical duration but only at 67% of the subjects' peak power output (67% W_{peak} , ~276 W). Exercise-induced peripheral locomotor muscle fatigue, as assessed via pre- and postexercise magnetic femoral nerve stimulation, was severe after the 83% W_{peak} trial and moderate after the 67% W_{peak} trial (Fig. 2). Now, on three additional days, all subjects performed 5-km cycling time trials during which they were able to voluntarily choose their power output to finish the task as fast as possible. The first time trial was performed in a “fresh” state (TT-Ctrl), i.e., without any preexisting fatigue. On the second day, subjects first repeated constant-load exercise (83% W_{peak}) to induce a severe level of preexisting locomotor muscle fatigue and then, after a 4-min break, performed a 5-km time trial (TT-severe). On the third day, subjects first repeated the 67% W_{peak} trial to induce a moderate level of preexisting locomotor muscle fatigue and then, after a 4-min break, performed a 5-km time trial (TT-moderate).

Preexisting locomotor muscle fatigue had a substantial dose-dependent inverse effect on CMD and power output during the 5-km time trials and a direct effect on performance time. Specifically, the higher the level of preexisting locomotor muscle fatigue, the lower the average CMD and power output during the subsequent time trial (Fig. 3). The

striking finding was that at the end of exercise, the level of peripheral fatigue was identical between the time trials— independent of the level of preexisting fatigue and/or the marked differences in exercise performance (Figs. 2 and 3) (3). For instance, the TT-severe time trial was started with a severe level of preexisting locomotor muscle fatigue as induced via high-intensity constant-workload exercise to exhaustion (83% W_{peak}). Hence, the individual critical threshold of peripheral fatigue and associated sensory tolerance limit had already been reached when the time trial started. Astonishingly, because the level of locomotor muscle fatigue at the end of the time trial was identical compared with the preexisting level at the start of the time trial (i.e., at the critical threshold) (Fig. 2), the subjects, who were instructed to finish the time trial as fast as possible, must have “chosen” CMD and associated power output throughout the race low enough to result in no further accumulation of peripheral fatigue (3). On the other hand, when the time trial was started with no preexisting fatigue (TT-Ctrl) or a lower level of preexisting

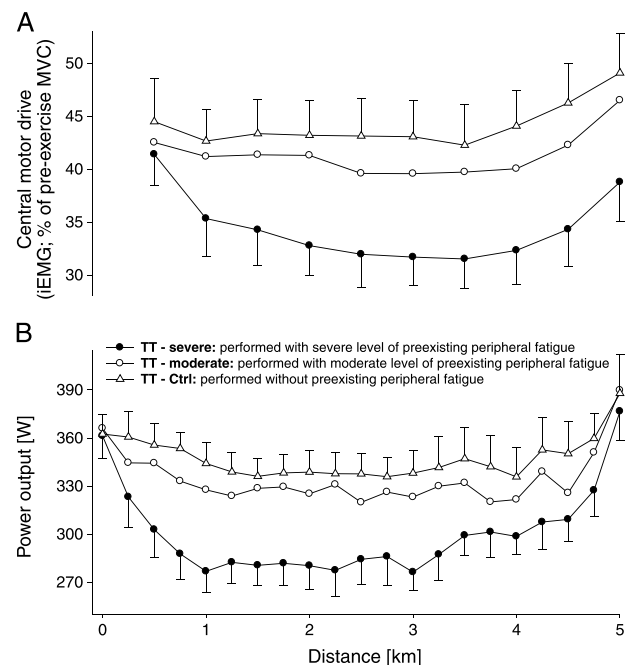


FIGURE 3—Effect of preexisting locomotor muscle fatigue on CMD and power output during a 5-km time trial. The control time trial (TT-Ctrl) was performed without preexisting locomotor muscle fatigue. The two experimental time trials were performed with different levels of preexisting quadriceps fatigue (percent reduction in quadriceps twitch force of about -36% and -20% for TT-severe and TT-moderate, respectively). A, Effects of preexisting locomotor muscle fatigue on group mean CMD (as estimated via integrated EMG (iEMG) of vastus lateralis normalized to the iEMG obtained during preexercise (unfatigued) maximal voluntary contractions (MVC) of the quadriceps). Each point represents the mean CMD of the preceding 0.5-km section. Mean CMD during the time trial was significantly reduced from TT-Ctrl to TT-severe. B, Group mean variations in power output during the 5-km time trial with three different levels of preexisting fatigue. Values of group mean power output / performance time were 347 ± 14 W / 7.3 ± 0.1 min, 298 ± 14 W / 7.8 ± 0.1 min, and 332 ± 18 W / 7.5 ± 0.1 min ($P < 0.05$) for TT-Ctrl, TT-severe, and TT-moderate, respectively. The subjects were required to reach an individual target power output before the race was launched. From Amann and Dempsey (3), used with permission.

locomotor muscle fatigue (TT-moderate), peripheral fatigue further accumulated throughout the subsequent time trial to reach the critical threshold at the end of exercise (Fig. 2) (3). With this study, we intended to directly challenge our hypothesis, and the outcome supported a crucial role of locomotor muscle fatigue on exercise performance via its inhibitory influence on CMD and, furthermore, confirmed the status of peripheral fatigue as a regulated variable.

However, a limitation is imposed on the interpretation of these findings because the prefatiguing exercise might also bring into play other nonperipheral effectors of central fatigue (3,31). Gagnon et al. (17) have recently tried to circumvent this limitation in a study including healthy subjects and patients with chronic obstructive pulmonary disease. These investigators used electrical stimulation of both rested quadriceps muscles to induce peripheral locomotor muscle fatigue without using voluntary muscle contractions and evaluated constant-load cycling exercise performance immediately after. They found that compared with control exercise, cycling time to exhaustion is significantly compromised when the identical constant-workload test is repeated with preinduced peripheral locomotor muscle fatigue. Despite these differences in exercise performance, the level of end-exercise peripheral locomotor muscle fatigue was similar in both trials. Taken together, their findings (17) not only supported but also nicely extended our results (3) and further confirmed our hypothesis.

After this first direct confirmation, we moved on to an even more specific intervention. Namely, we pharmacologically blocked sensory feedback from the fatiguing locomotor muscles and thus eliminated the inhibitory influence on CMD and the accompanying restriction of the development of peripheral fatigue during high-intensity whole-body endurance exercise. As a reminder, the key component of our proposed “regulatory mechanism” (Fig. 1) is the afferent arm consisting of both myelinated (group III) and unmyelinated (group IV) nerve fibers that increase their spontaneous discharge—and therefore their central projection—during exercise.

As a first step, we blocked the central projection of locomotor muscle afferent feedback during a 5-km cycling time trial via the lumbar epidural injection of a local anesthetic (0.5% lidocaine, vertebral interspace L3–L4) (8). However, lidocaine also affected efferent motor nerves leading to a significant loss in resting locomotor muscle strength (~22%). These confounding effects did not allow us to adequately test the role of afferent feedback effects *per se* on exercise performance. Indeed, power output during the time trial performed with the local anesthetic was lower as compared with the control trial. However, several lines of evidence were observed that support a higher CMD during the time trial performed with blocked locomotor muscle afferents. For example, EMG activity (relative to the maximal EMG measured during prerace maximal voluntary muscle contractions—which was lower with vs without epidural lidocaine) obtained from the vastus lateralis suggests that on average and over time, the “drive” to race averaged about 9% stronger when

neural feedback was blocked (8). Furthermore, cardiorespiratory variables (minute ventilation, HR, blood pressure) are known to reflect increases in CMD (11,40). A substantially increased CMD during the time trial with impaired neural feedback was reflected by the similar or even greater cardiovascular and respiratory response to exercise despite the significantly lower power output and metabolic rate during the lidocaine versus control time trial. In other words, HR and mean arterial blood pressure were nearly identical, and minute ventilation was even significantly increased despite the lower power output and metabolic rate during the lidocaine versus control 5-km time trial (8).

To circumvent the lidocaine-induced forfeit of locomotor muscle force-generating capacity and to adequately determine the effect of neural feedback from exercising muscle on power output and the development of peripheral fatigue during whole-body endurance exercise, we then used fentanyl (intrathecally, L3–L4), an opioid analgesic, to selectively block the central projection of ascending sensory pathways without affecting motor nerve activity or maximal force output (2,9). Again, the subjects had to perform a 5-km cycling time trial either with (fentanyl) or without (placebo);

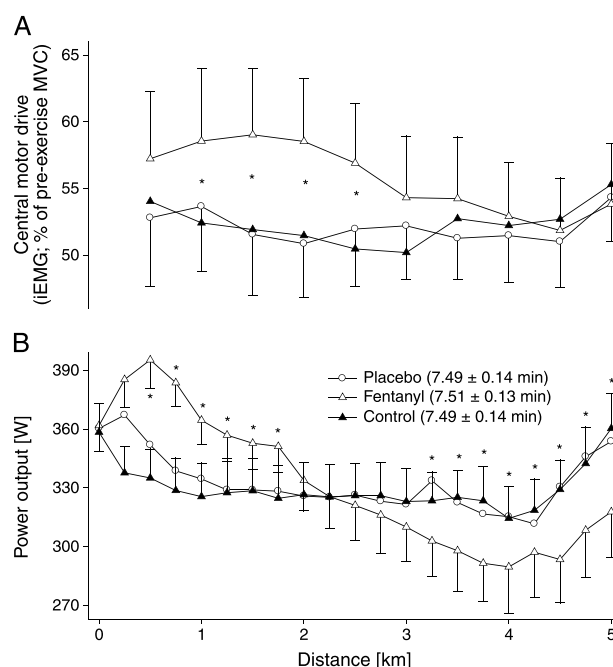


FIGURE 4—Effect of afferent blockade on CMD and power output during a 5-km cycling time trial. All subjects raced with no intervention (Control), with a placebo injection (Placebo; interspinous ligament injection of sterile normal saline, L3–L4), and with intrathecal fentanyl (Fentanyl, L3–L4). A, Effects of opioid analgesic (fentanyl) on group mean CMD as estimated via changes in iEMG of vastus lateralis. Mean iEMG of the vastus lateralis was normalized to the iEMG obtained from preexercise MVC maneuvers performed either without (Placebo and Control) or with (Fentanyl) intrathecal fentanyl. Each point represents the mean CMD of the preceding 0.5-km section. B, Group mean power output during the 5-km time trial with and without impaired afferent feedback. The subjects were required to reach an individual target power output before the race was launched. * $P < 0.05$ (Fentanyl vs Placebo). $N = 9$. From Amann et al. (9), used with permission.

intraspinal ligament injection of saline, L3–L4) opioid-mediated neural feedback from the locomotor muscles. Blocking these fibers attenuated the centrally mediated inhibitory effect, and CMD during the fentanyl time trial was less restricted and significantly higher as normally chosen by the athlete, i.e., in the placebo time trial (Fig. 4). This higher CMD resulted in a substantially higher power output during the first half of the race, and the CNS “allowed” or “tolerated” the exercise-induced development of peripheral locomotor muscle fatigue drastically beyond levels as observed after the same exercise but with an intact neural feedback system (Fig. 5) (9). In the absence of afferent feedback, the magnitude of CMD was thus uncoupled from the intramuscular metabolic milieu of the locomotor muscles. As a consequence, the “naïve” CNS did not limit the development of excessive peripheral fatigue beyond the individual critical threshold, which caused ambulatory problems like short-term difficulties with upright standing and walking. Nevertheless, the resulting metabolic and respiratory acidosis and the accompanying arterial hypoxemia (resulting from hypoventilation due to the missing afferent feedback [2]) facilitated a faster development of peripheral locomotor muscle fatigue and eventually prevented the performance to be improved during the fentanyl versus placebo time trial (9).

These last experiments also confirm the critical role of locomotor muscle afferents in regulating pacing strategy (14,38). When exercising with blocked group III/IV muscle afferent feedback, the athletes altered their pacing strategy and maintained a higher CMD throughout the race (Fig. 5). Although the overall exercise performance was, despite the higher CMD, unchanged from placebo conditions, a definite judgment of the newly adapted pacing strategy is difficult. This is because the missing afferent feedback also attenuated the ventilatory and circulatory response to exercise (which

facilitates the development of peripheral fatigue) (2,9), and this effect might have prevented the increased CMD to be reflected in improved time trial performance.

RELATIVE IMPORTANCE OF THIN-FIBER MUSCLE AFFERENTS IN DETERMINING CMD

Peripheral locomotor muscle fatigue and/or associated sensory feedback is only one of several potential mechanisms (31) influencing CMD and thus performance during high-intensity whole-body endurance exercise. However, this regulatory mechanism seems to influence the determination of CMD under normal conditions but might become secondary when exercise is performed under adverse physiological circumstances (e.g., hypoglycemia [31]) or mental stress/fatigue (27) or in the face of extreme environmental influences, such as heat (20,37) or severe hypoxia (6), which impose an immediate threat to the CNS of the exercising individual.

For example, we have shown that the relative effects of centrally versus peripherally originating impairments of CMD (and, consequently, exercise performance) change with the level of cerebral oxygenation (10). In a recent study, we instructed our subjects to exercise (bicycle) against a heavy-intensity fixed workload (333 ± 9 W) to exhaustion in normoxia (exercise time to exhaustion ~ 10 min, hemoglobin saturation at exhaustion $\sim 93\%$) and acute severe hypoxia (~ 2 min, $\sim 67\%$). When subjects stopped exercising at exhaustion in normoxia, peripheral locomotor muscle fatigue reached the individual critical threshold (10). In contrast, when the participants stopped exercising at exhaustion in severe hypoxia, peripheral muscle fatigue was significant but only about two-thirds of the level of fatigue measured at exhaustion in normoxia and therefore far below the individual critical threshold (10). In other words, subjects could have accumulated more fatigue, but they stopped exercising before their critical threshold was reached. Now, when we, similarly to Kayser et al. (25), surreptitiously switched the inspirate to a gas mixture with supplemental oxygen (30% O₂, hyperoxia) at exhaustion in normoxia (i.e., peripheral locomotor muscle fatigue has reached critical threshold), our subjects were not able to continue the exercise. In contrast, when we surreptitiously administered supplemental oxygen at exhaustion in severe hypoxia (i.e., peripheral locomotor muscle fatigue below critical threshold), all subjects were able to continue the exercise until they finally reached their critical threshold at exhaustion under hyperoxic conditions (10).

These findings clearly indicate the relative importance of our hypothesis. Although peripheral locomotor muscle fatigue and associated inhibitory feedback might be a major determinant of CMD under normal conditions, the relative importance of this inhibitory feedback on CMD seems to vanish in the face of a direct threat to the CNS, in this case, presumably severe cerebral hypoxemia, to the exercising individual. It seems that during exercise under extreme environmental conditions, other sources of inhibition of CMD

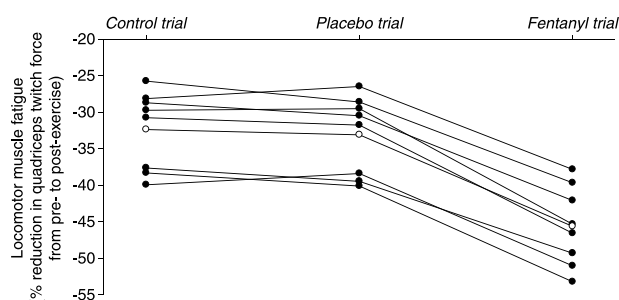


FIGURE 5—Individual (solid symbols) and group mean (open symbols) effects of 5-km time trial without (control and placebo trials) and with intrathecal fentanyl (fentanyl trial) on locomotor muscle fatigue expressed as a percent change in quadriceps twitch force (magnetic femoral nerve stimulation) from before to 3 min after exercise. Exercise performance was similar between control and placebo trials (~ 7.49 min, $P = 0.75$), which was also reflected in similar exercise-induced reductions in potentiated quadriceps twitch force from before to 3 min after the time trial. Despite a similar overall exercise performance (7.51 ± 0.13 min), end-exercise quadriceps fatigue was significantly exacerbated after the fentanyl versus placebo trial ($P < 0.001$). From Amann et al. (9), used with permission.

may outweigh the limiting effects of peripheral locomotor muscle fatigue and associated inhibitory feedback.

Finally, it seems that the group III/IV-mediated inhibitory feedback effects on CMD can be “ignored,” for a very brief period, by the exercising human. This statement stems from the observation that power output/speed at the end of and sometimes during a time trial often equals or even exceeds that observed at the beginning of the task (3,5,23). These short-term (30–60 s) increases in power output/speed evidence that the CNS is able to “override,” for a short period, the inhibitory feedback from muscle afferents and that it remains capable of briefly increasing CMD and thus speed—even in conditions of severe peripheral locomotor muscle fatigue (and associated inhibitory afferent feedback).

SUMMARY

We have been hypothesizing that exercise-induced alterations of the metabolic milieu (and associated peripheral fatigue) of locomotor muscles affect, in a dose-dependent manner, the firing rate—and thus the central projection—of muscle afferents providing inhibitory feedback to the deter-

mination of CMD during high-intensity whole-body endurance exercise. The purpose of this proposed feedback loop might be to regulate and restrict the level of exercise-induced peripheral locomotor muscle fatigue and/or the magnitude of sensory feedback to an “individual critical threshold.” This regulatory mechanism is relevant to strenuous endurance exercise under normal conditions, whereas under extreme environmental and/or physiological conditions, other sources of inhibition of CMD can outweigh the limiting effects of peripheral locomotor muscle fatigue and associated neural feedback.

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